Counting

Supantha Pandit

DA-IICT

Why counting?

- Want to count the total number of license plates made up of 7 numbers and alphabets?
- Want to count the total number of operations required for an algorithm?

Counting techniques

- Some basic counting techniques are as follows.
- In practice, a mix of these counting techniques is applied.

- The sum rule
- The product rule
- The complement rule
- Inclusion-exclusion principle
- Pigeon hole principle
- Permutations and combinations

The sum rule

Example: Suppose one has 8 shirts and 13 t-shirts. He/she has to attain a function. How many choices he/she has for an outfit?

Answer: $8 + 13 = 21$

The sum rule: If a task C can be done

either in one of n_1 ways of A

or in one of n_2 ways of B,

then there are $n_1 + n_2$ ways to do the task C.

Set theoretic notion: If A and B are disjoint sets, then $|A \cup B| = |A| + |B|$

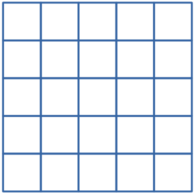
The sum rule: example

Example: In SC205 class there are 25 girls and 27 boys. If the instructor asks a question to a student. How many choices the instructor have in total?

Answer: $25 + 27 = 52$

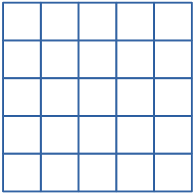
The sum rule: example

Example: We have the following figure. How many squares are there?



The sum rule: example

Example: We have the following figure. How many squares are there?



Answer: Let S_i be the number of squares of length i.

$S_1 = 25$ $S_2 = 16$ $S_3 = 9$ $S_4 = 4$ $S_5 = 1$

Total $= 55$

Generalization:

$S_1 = m \times n$

$S_2 = (m-1) \times (n-1)$

$S_3 = (m-2) \times (n-2)$

$S_i = (m-(i-1)) \times (n-(i-1))$

The sum rule: generalized

If $A_1, A_2, \ldots, A_n$ are disjoint events of a task A, then,
$|A| = |\cup_{i=1}^{n} A_i| = \sum_{i=1}^{n} |A_i|$

Questions: What if some of the tasks have intersections?
Important note: For sum rule to apply, the possibilities must be mutually exclusive (disjoint)

The product rule

Example: Suppose one has 8 shirts and 5 pants. He/she has to attain a function. How many choices he/she has for an outfit?
Answer: $8 \times 5 = 40$

The product rule

The product rule: If a task C can be broken down into two successive tasks A and B

if there are n_1 ways to do the first task A

for each way of doing the first task A, there are n_2 ways to do the second task B,

then there are $n_1 \cdot n_2$ ways to do the task C.

Set theoretic notion: If A and B are sets, then $|A \times B| = |A| \times |B|$

Generalized product rule: Suppose task A decomposes into successive tasks $A_1, A_2, \ldots, A_k$

if there are n_1 ways of doing A_1, ..., n_k ways of doing A_k,

then there are $n_1 \cdot n_2 \cdot \ldots \cdot n_k$ ways of doing A

The product rule: example

Example: How many ways can we arrange the two letters A and B?
Answer: 2 ways

- choose the first letter (2 ways)
- choose the second letter (for each first choice 1 way)
- Total: $2 \times 1 = 2$ ways; AB and BA

Example: How many ways the letters A, B, C, D can be arranged?
Answer:

- choose the first letter (4 ways)
- choose the second letter (3 ways)
- choose the third letter (2 ways)
- choose the forth letter (1 way)
- Total: $4 \times 3 \times 2 \times 1 = 24$ ways

In general:

Number of ways to arrange n letters is $n!$.

The product rule: example

Example: there are 4 boys and 3 girls. A team is to form of 2 containing 1 boy and 1 girl. How many different teams can be possible?

Answer: $4 \times 3 = 12$

- choose a boy (4 ways)
- choose a girl (for each boy there are 3 ways)

Example: A license plate is made that contains 3 letters and 4 digits. The first 3 are letters and the next 4 are digits. How many plates can be possible?

Answer: When the letters and digits are repeated
$26 \times 26 \times 26 \times 10 \times 10 \times 10 \times 10$
When the letters are not repeated $26 \times 25 \times 24 \times 10 \times 10 \times 10 \times 10$

Example: mix of sum, product rules

Example: A code is generated. It contains digits and letters (a-z and A-Z). the length of the code can be 5 to 8 character long. It must begin with a letter. repetation is allowed.
How many codes can be possible?

Answer: $N_i =$ number of codes of length i
We want to compute: $N_5 + N_6 + N_7 + N_8$ (sum rule)
Computing each N_i (using product rule)
$N_5 = 52 \times 62 \times 62 \times 62 \times 62$
$N_6 = 52 \times 62 \times 62 \times 62 \times 62 \times 62$
$N_7 = 52 \times 62 \times 62 \times 62 \times 62 \times 62 \times 62$
$N_8 = 52 \times 62 \times 62 \times 62 \times 62 \times 62 \times 62 \times 62$

The complement rule

Example: Suppose one wants to roll two dice. How many possibilities that the two dice shows different values?

Answer: Number of possibilities in the first dice (6)

For each choice of the first dice number of possibilities in the second dice (5)

Total: 30

Using complement rule:
Possibilities = total number - two dice show the same value
$= 6 \times 6 - 6 = 36 - 6 = 30$

More complex example

Example: How many bit strings are there of length 6 that do not have two consecutive 1's?

Answer: Let $S(n)$ denote the number of bit strings of length n that do not have two consecutive 1's

We'll first derive a recursive equation to characterize $S(n)$

By the sum rule, $S(n)$ is the sum of:

T1: # of n-bit strings starting with 1 not containing 11

T2: # of n-bit strings starting with 0 not containing 11

Our target is to compute T1 and T2 and then compute T1 $+$ T2.

Computing T1

How many n-bit strings are there starting with 1 and not containing 11?

Since we do not want two consecutive 1's, the second bit must be a 0

But the rest can be anything as long as it doesn't contain two consecutive 1's

Thus, the rest = number of bit strings of length $n-2$ not containing two consecutive 1'

But this is precisely $S(n-2)$

Thus, by the product rule, there are $1 \cdot 1 \cdot S(n-2) = S(n-2)$ n-bit strings starting with 1 and not containing 11

Computing T2

How many n-bit strings are there starting with 0 and not containing 11?

Since the first bit is a 0, the rest can be anything as long as it doesn't contain two consecutive 1's

Thus, the rest = number of bit strings of length $n-1$ not containing two consecutive 1'

But this is precisely $S(n-1)$

Thus, by product rule, there are $1 \cdot S(n-1) = S(n-1)$ n-bit strings starting with 1 and not containing 11

Computing total number

Recall that by the sum rule, $S(n)$ is the sum of:

T1: # of n-bit strings starting with 1 not containing 11

T2: # of n-bit strings starting with 0 not containing 11

Our target is to compute T1 and T2 and then compute T1 + T2.

Calculating T1: $S(n-2)$ Calculating T2: $S(n-1)$

Therefore, $S(n) = s(n-1) + S(n-2)$

This is a recursive definition, but what are the base cases?

What is $S(1)$? What is $S(2)?$

Original question asks for $S(6)$

Using base cases and recursive definition:

What is $S(3)?$ What is $S(4)?$ What is $S(5)?$ What is $S(6)?$

The complement rule

The complement rule: To count the element of a set A, find the element in the set $\overline{A}$. Then $|A| = |U| - |\overline{A}|$

Example: Suppose a code of length 5 need to be generated

How many possibilities are there if at least one repetition is allowed?

Answer: $10^5 - 10 \times 9 \times 8 \times 7 \times 6$

All possible cases $-$ number of no repetition

$10^5 - 10 \times 9 \times 8 \times 7 \times 6$

Inclusion-exclusion principle

- **Recall:** Sum rule only applies if a task is as disjunction of two mutually exclusive tasks.
- What do we do if the tasks are not mutually exclusive?
- **Example:** You can choose from the set A or the set B, but they have some elements in common.
- In such cases, the sum rule is not applicable because common elements are counted twice
- Fortunately, there is a generalization of the sum rule called the **inclusion-exclusion** principle

Inclusion-exclusion principle

Example: There are 50 boys and 30 girls in MC123. The instructor wants to give a grade $Pass$ to a student. How many ways can the instructor do it?

Answer: $50 + 30 = 80$

Example: There are 50 students in MC123 and 30 students in MC110. An instructor teaching both courses MC123 and MC110. The instructor wants to give a grade $Pass$ to a student. How many ways can the instructor do it?

Answer: $50 + 30 = 80$ - - - - Not always true.

Reason: Some students may take both courses MC123 and MC110.

Inclusion-exclusion principle

Two sets:
$|A \cup B| = |A| + |B| - |A \cap B|$

Three sets:

$|A \cup B \cup C| = |A| + |B| + |C| - |A \cap B| - |B \cap C| - |A \cap C| + |A \cap B \cap C|$

In general:

$$|\cup_{i=1}^{n} A_i| = \sum_{i=1}^{n} |A_i| - \sum_{1 \le i < j \le n} |A_i \cap A_j| + \sum_{1 \le i < j < k \le n} |A_i \cap A_j \cap A_k| - \ldots + (-1)^{n-1} |A_1 \cap A_2 \cap \ldots \cap A_n|$$

Inclusion-exclusion principle

Example: Consider an integer between 1 and 100. How many of them are divisible by either 7 or 9

Answer: Total numbers = numbers multiple of 7 + numbers multiple of 9 - numbers multiple of their common integers
$\lfloor \frac{100}{7} \rfloor + \lfloor \frac{100}{9} \rfloor - \lfloor \frac{100}{63} \rfloor$

The pigeon hole principle

- Suppose there are some pigeons and some holes.
- the number of pigeons is more than the number of holes.
- Then some pigeon holes must have more than one pigeon.

Example: there is a set of 5 cards. Can we say that at least 2 of the cards have the same suit?

Answer: yes

The Pigeonhole Principle: If $n+1$ or more pigeons are placed into n holes, then at least one hole contains 2 or more pigeons.

Generalized pigeon hole principle

Example: Any set of 13 cards drawn from a deck of cards. How many of them are of same suit?

Answer: $\geq \lceil \frac{13}{4} \rceil$, i.e., at least 4 cards.

Generalized Pigeonhole Principle: If n pigeons are placed into h holes, then there is at least one hole containing at least $\lceil \frac{n}{h} \rceil$ pigeons.

proof: (by contradiction) Suppose every box contains less than $\lceil \frac{n}{h} \rceil$ objects.

Then, there must be less than $h \cdot (\lceil \frac{n}{h} \rceil - 1)$ pigeons.

We already know that $\lceil \frac{n}{h} \rceil < \lceil \frac{n}{h} \rceil + 1$

Adding one and multiply both sides by h: $h \cdot (\lceil \frac{n}{h} \rceil - 1) < n$

Thus, this would imply that there are fewer than n objects, a contradiction.

Generalized pigeon hole principle

Example: Suppose there are 250 lacks phones in a state.
Each phone number has 10 digits, where the first digit is $[6-9]$ and other digits are $[0-9]$.
What is the minimum number of area codes necessary to guarantee each phone has a unique number?

Answer: First, compute the number of 10-digit phone numbers:
Let n be the number of area codes.
Want to find a minimum n such that
$\lceil \frac{250\times10^5}{n} \rceil \leq 4\times10^9$

Permutations

Example: How many arrangements are there for 3 letters A, B, C.

Answer: Decompose using the product rule.

- choose the first letter (3 ways)
- choose the second letter (2 ways)
- choose the third letter (1 way)
- Total $3 \times 2 \times 1 = 6$ ways

For n letters it is $n \times (n-1) \times \ldots \times 1 = n!$ ways

Example: What is the number of 3 letter codes form a set of 6 characters set and no repetition.

Answer: Total number of codes: $6 \times 5 \times 4 = 120$

r digits code from a set of n characters set and no repetition is $n \times (n-1) \times \ldots \times (n-(r-1))$.

Permutations

Permutation: A permutation or arrangement of a set of n distinct objects is an ordering of the objects.

- No object can be selected more than once
- order of arrangement matters
- The number of permutations of n objects is $n!$.

Generalized permutation: A r-permutation or r-arrangement of a set S of n objects is an ordering of r objects out of n objects.

- S contains more than r objects.
- The number of r-permutations of n objects is $P(n, r) = n \times (n-1) \times \ldots \times (n-(r-1)) = \frac{n!}{(n-r)!}$.

Ordered vs unordered arrangements

Example: There are 6 students. A grade AA will be assigned to 3 students. How many choices are possible?

Ordered vs unordered arrangements

Example: There are 6 students. A grade AA will be assigned to 3 students. How many choices are possible?

Answer:

Fix	Fix	Forced	count
A	B	C,D,E,F	4
A	C	D,E,F	3
A	D	E,F	2
A	E	F	1
B	C	D,E,F	3

Fix	Fix	Forced	count
B	D	E, F	2
B	E	F	1
C	D	E,F	2
C	E	F	1
D	E	F	1

Total $4+3+2+1+3+2+1+2+1+1=20$

Ordered vs unordered arrangements

Example: There are 6 students. A grade AA will be assigned to 3 students. How many choices are possible?

Answer:

Fix	Fix	Forced	count
A	B	C,D,E,F	4
A	C	D,E,F	3
A	D	E,F	2
A	E	F	1
B	C	D,E,F	3

Fix	Fix	Forced	count
B	D	E, F	2
B	E	F	1
C	D	E,F	2
C	E	F	1
D	E	F	1

Total $4+3+2+1+3+2+1+2+1+1=20$

Idea: ABC, ACB, BAC, BCA, CAB, CBA receive same grades AA

- The number of 3 student arrangements from the set $\{A, B, C, D, E\}$ is $P(6,3)$
- Each unordered 3 students counted 6 times, i.e., $3!$ times
- Total possibilities $\frac{P(6,3)}{3!} = \frac{6!}{3!3!} = 20$

Ordered vs unordered arrangements

Example: There are 6 students. Grades AA, BB, CC will be assigned to 3 students. How many choices are possible?

Ordered vs unordered arrangements

Example: There are 6 students. Grades AA, BB, CC will be assigned to 3 students. How many choices are possible?

Answer: In this case the order matters.

ABC is not the same as BAC

The total of possibilities is $P(6,3) = 120$

Ordered vs unordered arrangements

Example: Consider a set $S = \{0, 1, 2, 3, 4, 5, 6, 7, 8, 9\}$. Count the number of subsets such that each subset contains 5 elements from S.

Ordered vs unordered arrangements

Example: Consider a set $S = \{0, 1, 2, 3, 4, 5, 6, 7, 8, 9\}$. Count the number of subsets such that each subset contains 5 elements from S.

Answer:

- The number of subsets where order matters is $P(10, 5)$.
- The number of arrangements/ordering of 5 elements is $P(5, 5)$.
- each subset is counted $5!$ times
- The total number of (unordered) subsets is $\frac{P(10,5)}{5!}$

Here the concept of combinations comes!

Ordered: Permutations

Unordered: Combinations

Combinations

Combination: A r-combination is a (unordered) subset of size r of n objects is

$$C(n,r) = \frac{P(n,r)}{r!} = \frac{n!}{(n-r)!r!}$$

$C(n,r)$ also denoted as $\binom{n}{r}$ (binomial coefficient)

Example: We have a set $S = \{a,b,c,d\}$.

Number of 0-subsets of $S = \binom{4}{0} = \frac{4!}{(4-0)!0!} = 1$

Number of 1-subsets of $S = \binom{4}{1} = \frac{4!}{(4-1)!1!} = 4$

Number of 2-subsets of $S = \binom{4}{2} = \frac{4!}{(4-2)!2!} = 6$

Number of 3-subsets of $S = \binom{4}{3} = \frac{4!}{(4-3)!3!} = 4$

Number of 4-subsets of $S = \binom{4}{4} = \frac{4!}{(4-4)!4!} = 1$

Combinations

Example: We have a sequence of 3 bit numbers. How many sequences are possible that contain 2 0's and 1 1's?

Answer:

- 3 positions to put a 0
- 2 positions to put a 0
- 1 positions to put a 1
- ordered case $3 \times 2 \times 1 = 6$
- unordered case: $\frac{6}{2!} = 3$

Two positions are filled with 0 and the remaining one position is forced to fill a 1.
$\binom{3}{2} = 3$

Combinations

Example: A committee will be formed with 10 people. It has 7 girls and 3 boys. There are 20 girls and 15 boys. How many different committees can be formed?

Answer: $\binom{20}{7} \times \binom{15}{3}$

Example: There is a 10 bit sequence. How many sequences is possible with 4 0's and 6 1's?

Answer: $\binom{10}{4}$ or $\binom{10}{6}$

Binomial identities

$C(n,r) = \frac{P(n,r)}{r!} = \frac{n!}{(n-r)!r!}$

$C(n,r)$ also denoted as $\binom{n}{r}$ (binomial coefficient)

Properties:

- $\binom{n}{r} = \binom{n}{n-r}$
  Selecting r objects from n objects $\Leftrightarrow$ Rejecting $n-r$ objects from n objects.
  Algebraic proof: $\binom{n}{n-r} = \frac{n!}{(n-(n-r))!(n-r)!} = \frac{n!}{r!(n-r)!} = \binom{n}{r}$
- $r\binom{n}{r} = n\binom{n-1}{r-1}$
  Algebraic proof:
  $r\binom{n}{r} = r\frac{n!}{(n-r)!r!} = \frac{n!}{(n-r)!(r-1)!} = \frac{n(n-1)!}{((n-1)-(r-1))!(r-1)!} = n\binom{n-1}{r-1}$
- **Pascal identity:** $\binom{n}{r} = \binom{n-1}{r} + \binom{n-1}{r-1}$
  proof: HW

Binomial identities: properties

pascal identity:
$\binom{n}{r} = \binom{n-1}{r} + \binom{n-1}{r-1}$
proof: HW

Permutations with repetition

Question: How many r-permutations with repetitions of n objects are possible?
Answer: n^r
Example: How many different license plates are possible?
7 letters long and first 4 letters are alphabet.

26	26	26	26	10	10	10

Combinations with repetitions

Example: There are three different types of fruits, 20 apples (A), 30 bananas (B) and 40 cherries (C) to choose from.
How many ways can 4 fruits be selected?

A	A	A	A	A	B	C	C
A	A	A	B	A	C	C	C
A	A	A	C	B	B	B	B
A	A	B	B	B	B	B	C
A	A	B	C	B	B	C	C
A	A	C	C	B	C	C	C
A	B	B	B	C	C	C	C
A	B	B	C				

Total Possibilities: 15.

In-general: Number of r combinations with repetitions from n-element set. $\binom{n+r-1}{r}$

In this example: $\binom{3+4-1}{4} = \binom{6}{4} = 15$

Example

Example: 7 different types of rupee notes are there: 100, 50, 20 10, 5, 2, 1
Select 5 notes.
How many possibilities are there?
Assumption: there are at least 5 notes for each type.
Solution:
The star-bar method:
$**|||**|*||$ $\Rightarrow$ there are 2 100, 0 50, 0 20, 2 5, 1 2, 0 1 rupee notes.
We need to arrange these 6 dividers and 5 stars.
Number of ways to select the positions of 5 stars from the 11 positions.
The answer is $\binom{11}{5}$

Combinations with repetitions

There are $\binom{n+r-1}{r}$, r-combinations from a set with n elements with repetitions.

Example: There is a company produced 6 different types of products. In how many ways 7 products can be selected?

Solution: $\binom{6+7}{7} = \binom{12}{7}$